How Much Do Families Dislike School Lotteries? Evidence from SFUSD's TK Promotion Policy

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Abstract

Popular media suggests that school lotteries are unpopular with families. We leverage San Francisco Unified School District's (SFUSD) transitional kindergarten (TK) promotion priority—under which the 2024–25 TK cohort was offered a guaranteed kindergarten seat at its own school for 2025–26, in exchange for leaving the kindergarten lottery—to quantify families' distaste for lottery participation. Counting promoted families as having ranked their own school first, schools with TK programs received relatively more first-choice requests than schools without; observed first-choice requests, which exclude those families, instead fell by 27%. Building on a revealed-preference model of school value, we show that distaste for lottery participation is roughly equivalent to the quality difference between the best and median schools in the district. Because aggregate data cannot separate this distaste from selection on match value, we read the estimate as an upper bound, and set out what individual-level choice data would identify.

1 Introduction

Assignment lotteries are a common mechanism for allocating public school seats: as of 2022, 71 of the 300 largest school districts in the United States used a centralized-admission lottery ([4] and references therein). While assignment lotteries are believed to have a salutary effect on district enrollment, the popular conception is that families dislike them. For example, changes to Chile's *Sistema de Admisión Escolar* were covered with a newspaper illustration of a child strapped to a roulette wheel [3]; and in SFUSD's own fall 2026 survey of families, 82% chose a guaranteed spot near home when asked to pick between predictability and choice.[1]

Apparent distaste for assignment lotteries has combined with the ambiguous ability of lotteries to achieve their stated aims of improving student academic outcomes and decreasing segregation to yield a policy environment in which assignment lotteries are a target for de-implementation (typically in favor of traditional neighborhood schooling). If assignment lotteries are fulfilling no social objectives, and are scaring families

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[1]SFUSD, *Fall 2026 Family Survey and Focus Groups on SFUSD's Student Assignment Policy*, `https://www.sfusd.edu/school-reorganization/fall-2026-family-survey-and-focus-groups-sfusds-student-assignment-policy`. The survey drew 2,978 responses (1,871 SFUSD families, 1,107 not in SFUSD) between 11 September and 9 October 2026. It is not a random sample, and respondents skew White and higher-income relative to SFUSD families; the district reports that reweighting to neighborhood and income moves this figure by no more than three points.

away from public schools, then it stands to reason that they should be eliminated. In San Francisco, for example, school funding is determined by measured enrollment. A mechanism change that reduced San Francisco's private schooling rate from 30% to the national average of 9% would increase the system's LCFF revenue by roughly $136 million annually.[2]

To address the idea that a change to the school assignment system may improve enrollment, it is necessary to understand families' distaste for lottery participation. San Francisco's recent changes to its assignment lottery priorities provide a natural context in which to evaluate the magnitude of this distaste. San Francisco's school lottery is run as deferred acceptance (to a first approximation): students apply to schools in rank order, and each school has a priority order over students. The district computes a market priority for each school such that students who applied and are above this priority are admitted, and other students are rejected (cite the simple descriptions paper). The district's priority classes are coarse. In the 2023–24 school year SFUSD expanded its transitional kindergarten offerings and, to encourage enrollment, offered TK students top priority for kindergarten enrollment at their TK's associated elementary school. As a practical matter, all TK students who wished to remain at their school were allowed to do so.

At the same time, schools in SFUSD are commonly known to have heterogeneous qualities, based on parent beliefs. This, together with the fact that not all kindergarten sites have associated TK programs, suggests an empirical test. TK seats are scarce—there are not enough for every family that wants one—so TK enrollment is itself the outcome of a choice, a point we return to below. On the reading we test, a family that remains at a "low-quality" elementary school after TK reveals a distaste for lottery participation large enough to outweigh the quality it gives up, while a family that leaves a "high-quality" elementary school for the lottery reveals the opposite. Both readings require that TK families value their school as the city does. Section 4 shows they need not, and that aggregate data cannot separate distaste for the lottery from a taste for one's own school; that limitation is the central argument of this paper. By comparing the lottery participation rate of TK families to the measured quality of their associated elementary school, we can estimate the magnitude of families' distaste for lottery participation.

We first show that the TK promotion policy sharply reduced the number of families ranking kindergartens with TK programs first, and that counting promoted families as having ranked their own school first more than offsets that fall. This is consistent with both selection into TK programs and a distaste for lottery participation. We then estimate a revealed-preference model of school quality.[3] Next, we build a model of lottery participation in which a family may accept its current school or enter a costly lottery over all schools. We find that the cost of the lottery is roughly equivalent to the quality difference between the best and median schools in SFUSD. Because the same take-up pattern is equally consistent with families valuing their own school above the citywide average, we read this as an upper bound on distaste, and devote §3.4 to what individual-level choice data would identify.

[2]Own calculation; the chain is in `lcff_counterfactual.py` and `output/lcff_counterfactual.csv`. SFUSD enrolls 48,736 students in its own (non-charter) schools, and San Francisco's private schools about 23,765, or 30% of the city's K–12 students; the national private share is 9% (NCES). Moving to that share returns 14,621 students to the district—13,705 in average daily attendance, after SFUSD's attendance ratio, and after allocating the gain between the district and charter schools in current proportions. Under the Local Control Funding Formula these students add the base grant, but, not being unduplicated pupils, they dilute the concentration grant SFUSD draws because its unduplicated pupil percentage (60.4%) exceeds the 55% threshold: an influx this size takes it to 46.5%, below the threshold, so the concentration grant is lost on existing students too. Hence $136 million rather than the $190 million that SFUSD's current $13,887 per ADA would imply. The same formula reproduces SFUSD's reported LCFF per ADA to within 7%, which is the check on it. This is revenue, not net of the cost of educating the additional students.

[3]The approach follows [2], who show that a collective ranking constructed from submitted rank-order lists converges to the true quality ranking as the market grows large.

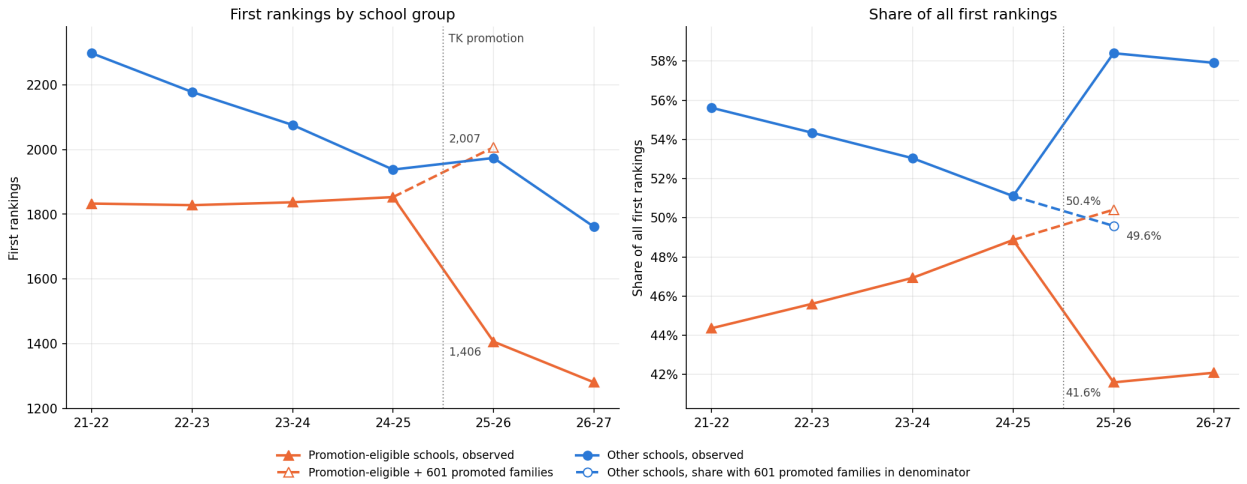


Figure 1: First-choice requests by school group, 2021–22 through 2026–27. Left panel: counts of first-choice requests at promotion-eligible schools and at other schools, with a hypothetical series that adds the 601 promoted families to eligible schools in 2025–26, as if each had ranked its own school first. Right panel: the same series as shares of all first-choice requests. Promotion sharply cuts observed first choices at eligible schools, and the hypothetical series more than restores them, lifting the eligible share slightly above its 2024–25 level. Source: `output/first_rankings_by_year.csv`.

1.1 Literature

2 Data

SFUSD provides aggregated ranking data for its annual assignment lottery. While the lottery places students for any school year, the main participation points (cite) are kindergarten, 6th grade, and 9th grade. We focus solely on kindergarten applications, as these applications constitute a large and independent market.

The lottery data are available for 15 application cycles, covering applications for the 2012–13 academic year through applications for the 2026–27 academic year. For each cycle, the data report the number of families who ranked each kindergarten program at each rank, and also the total number of families who applied. Ranks are reported individually only for the first five to nine choices, with deeper ranks pooled into a top-coded bucket, and the scheme changes across cycles: 2012–13 through 2018–19 report ranks 1–7 and then eighth-or-deeper; 2019–20 and 2020–21 report ranks 1–5, 6–10, and two deeper buckets; 2022–23 onward report ranks 1–9 and then tenth-or-deeper; and 2021–22 carries both schemes. List lengths are therefore variable and observed only coarsely beyond the ninth rank. The data also report the number of families who accepted automatic promotion from TK to kindergarten at each TK program in 2025–26.

We do not at present have access to individual-level data on families' rank lists, home locations, or TK enrollment and promotion decisions. We have requested these data from SFUSD via public records request, and will update the analysis once they are received. Without these data we cannot assess characteristic-based selection into TK programs.

3 Model

We apply a random-utility (logit) model of school preferences, written at the level of the individual family so that it can be taken directly to choice-level data (rank-order lists, home locations, TK enrollment and promotion decisions) once we obtain them via public records request. With only school-by-year request counts in hand today, every family-level term below is empty and the model collapses to a school fixed effect; we write it out in full so that the aggregate estimates in this draft are visibly a special case of the model we eventually want to estimate.

3.1 Preferences

Index families by i, kindergarten programs by $k \in \mathcal{K}_t$, and application cycles by t. Family i's utility from attending k is

$$u_{ikt} = \underbrace{\delta_k}_{\text{school FE}} + \underbrace{x_i'\beta_k}_{\substack{\text{student} \\ \text{characteristics}}} + \underbrace{z_{ik}'\gamma}_{\substack{\text{(student, school)} \\ \text{characteristics}}} + \varepsilon_{ikt},$$

where

- δ_k is the school fixed effect: the common component of every family's valuation of k (what the rest of this draft calls θ_k);
- x_i are exogenous family characteristics (race/ethnicity, home language and EL status, FRPM eligibility, CTIP1 status, TK enrollment). Because x_i does not vary across schools, it enters only through school-specific (or school-attribute-specific) slopes β_k—e.g. $x_i'\beta_k = \sum_m x_i' B\, w_{km}$ for observed school attributes w_{km} (immersion language, test scores, demographic composition)—so that, e.g., Spanish-speaking families can value Spanish immersion differently;
- z_{ik} are exogenous (family, school) characteristics: commute distance from home to k, whether k is i's attendance-area school, whether i has an older sibling enrolled at k, whether k is i's TK site. Distance is the term that will eventually let us convert utility into dollars;
- ε_{ikt} is i.i.d. Type-I extreme value with unit scale, which normalizes the units of every other term.

The outside option—not applying to a school, which includes private school, charter, leaving San Francisco, or simply ranking fewer schools—has utility

$$u_{i0t} = \xi_t + x_i'\beta_0 + \varepsilon_{i0t},$$

where ξ_t is a year effect absorbing citywide shifts in the attractiveness of SFUSD (e.g. the post-2020 enrollment decline). Only differences in utility are identified, so we normalize $\beta_0 = 0$ and one year's ξ_t to zero.

3.2 Applications

SFUSD's assignment mechanism is a variant of deferred acceptance, which is strategy-proof for families when rank lists are unrestricted, so we take a family's submitted rank list to be its true preference order over the schools it finds acceptable. Family i ranks k if and only if $u_{ikt} \geq u_{i0t}$, and orders the ranked schools by u_{ikt}.

The likelihood of an observed list $k_1 \succ k_2 \succ \cdots \succ k_L \succ 0$ is the rank-ordered ("exploded") logit,

$$\Pr\left(k_1 \succ \cdots \succ k_L \succ 0\right) = \prod_{\ell=1}^{L} \frac{\exp v_{ik_\ell t}}{\exp v_{i0t} + \sum_{m \geq \ell} \exp v_{ik_m t} + \sum_{j \notin \{k_1,\ldots,k_L\}} \exp v_{ijt}} \times \frac{\exp v_{i0t}}{\exp v_{i0t} + \sum_{j \notin \{k_1,\ldots,k_L\}} \exp v_{ijt}},$$

with $v_{ikt} = u_{ikt} - \varepsilon_{ikt}$ the deterministic part. With individual lists, $(\delta, \beta, \gamma, \xi)$ are estimated by maximum likelihood.

3.3 TK automatic promotion

A family i whose child attended TK at school $k(i)$ in 2024–25 was offered a guaranteed kindergarten seat at $k(i)$, conditional on not entering the lottery. Accepting yields

$$U_i^{\text{accept}} = v_{ik(i)t} + c + \eta_{i1},$$

while declining and entering the lottery yields the expected value of the lottery assignment,

$$U_i^{\text{lottery}} = \underbrace{\sum_{\ell} v_{ik_\ell t}\, P_{ik_\ell} \prod_{m<\ell} (1 - P_{ik_m})}_{\equiv \text{EU}_i} + \eta_{i0},$$

where P_{ik} is i's admission probability at k (a function of SFUSD's priority categories, which are themselves functions of z_{ik} and x_i), the sum runs over i's rank list, and η are i.i.d. Type-I extreme value. The parameter of interest is c: the utility a family will give up to avoid the lottery, i.e. its distaste for lottery participation (equivalently, a certainty premium). Then

$$\Pr\left(\text{accept}_i\right) = \Lambda\left(v_{ik(i)t} - \text{EU}_i + c\right),$$

with Λ the logistic CDF. Because the acceptance decision is a binary logit with the same utility index, c is measured in the same units as δ_k, and (once γ_{distance} is estimated) in dollars.

3.4 What survives without choice data

With only aggregate counts we must set $\beta = 0$ and $\gamma = 0$. Then $v_{ikt} = \delta_k$, $v_{i0t} = \xi_t$, every family is identical up to ε, and the probability any given applicant ranks k is $\exp\delta_k/(\exp\delta_k + \exp\xi_t)$.[4] Writing $\theta_k \equiv \delta_k$ and $\theta_{0t} \equiv \xi_t$ to match the rest of the draft, the number of applications to school k is

$$A_{kt} = N_t \times \frac{\exp\theta_k}{\exp\theta_k + \exp\theta_{0t}} \quad \Longrightarrow \quad \ln A_{kt} - \ln\left[N_t - A_{kt}\right] = \theta_k - \theta_{0t},$$

which we estimate by OLS of the log-odds $\ln A_{kt} - \ln(N_t - A_{kt})$ on school and year fixed effects. (The log-odds are not well approximated by the log-share $\ln A_{kt} - \ln N_t$: the most popular schools draw up to half of a year's applicants, e.g. Clarendon ES in 2017–18.) Wherever 2025–26 data are used, both A_{kt} and N_t add back the families who accepted automatic promotion, reconstructing applications as if the opt-out did not exist.

[4] This treats "rank k vs. not" as a binary comparison of k with the outside option, which is what the rank-ordered logit implies for the marginal event $\{u_{ik} \geq u_{i0}\}$.

Estimation window. Our lottery panel covers 15 application cycles, 2012–13 through 2026–27, but the main specification estimates θ_k from the four cycles 2021–22 through 2024–25 only. These are the cycles families deciding on promotion (in early 2025) had observed; they exclude the post-decision 2026–27 cycle and the treatment year itself; and their request volume matches the CDE enrollment years (2023–24, 2024–25) we use for seat capacity, whereas the 2015–18 cycles saw more than twice today's requests (about 50,000 ranked requests a year against about 20,000 now). This window covers 71 kindergarten programs. Section 5 shows the results are insensitive to the window and to measuring quality year by year.

Schools versus programs (a data limitation). The lottery reports requests by program, and the promotion roster lists eligibility and acceptance by TK program, but CDE reports kindergarten enrollment (our capacity measure) only by school. We therefore estimate quality, admission probabilities, and promotion rates at the school (building) level. Two promotion-eligible buildings run two TK programs each—Chinese Immersion School at DeAvila (two Chinese-immersion strands) and Dolores Huerta (two Spanish strands)—so we sum their roster rows to the building before merging with the lottery panel, and both strands share one θ_k and one admission probability. This merge is forced by the capacity data rather than a modeling choice: if demand differs between the strands of a building, we cannot see it. (An earlier version of the pipeline merged the roster program by program, which double-counted these two schools in every year of the panel; fixing this moved no reported estimate by more than 0.01.) Similarly, admission probabilities become school-level, $P_{ik} = P_k$, $\mathrm{EU}_i = \mathrm{EU}$ for every family, and the promotion decision becomes

$$\Pr\left(\text{accept} \mid k\right) = \Lambda\left(\theta_k - \mathrm{EU} + c\right),$$

which we take to the school-level promotion counts in §4. Everything the individual-level data add enters through x_i, z_{ik}, and individual P_{ik}; in particular, z_{ik} (distance, attendance area, sibling) is what lets the model distinguish "this family prefers its TK school" from "this family dislikes the lottery," which is the main selection concern below.

Observation. The number of applicants N_t is endogenous to the outside option θ_{0t}. However, the average applicant applies to about 4.84 schools; IIA in the logit model implies that, empirically, N_t should be effectively independent of θ_{0t}.

Data requested. For each kindergarten applicant (and each 2024–25 TK student) in 2012–13 through 2026–27, at the program level: an anonymized ID; full submitted rank list; home census block or attendance area; CTIP1 status; sibling priority by school; TK school and promotion offer/decision; assignment outcome; and October enrollment (to see who left SFUSD). Optionally: race/ethnicity, home language, FRPM status. Also: 2023–24 TK applications and rank lists, since where a 2024–25 TK family ranked its TK school reveals its own taste for that school; and kindergarten seat capacity by program.

TK-promotion eligibility appears roughly unrelated to measured school quality.

3.5 Effect of promotion on applications

A mechanical model. Let N_t be the number of lottery applicants in cycle t, P_{kt} the number of families who accepted promotion at TK school k, and $P_t = \sum_k P_{kt}$, so that absent promotion the cohort would have been $N_t + P_t$ families. A general applicant ranks school k with probability s_{kt} (under the logit model of §3, $s_{kt} = \exp\theta_k / (\exp\theta_k + \exp\theta_{0t})$). A promoted family from school j would, absent promotion, have ranked

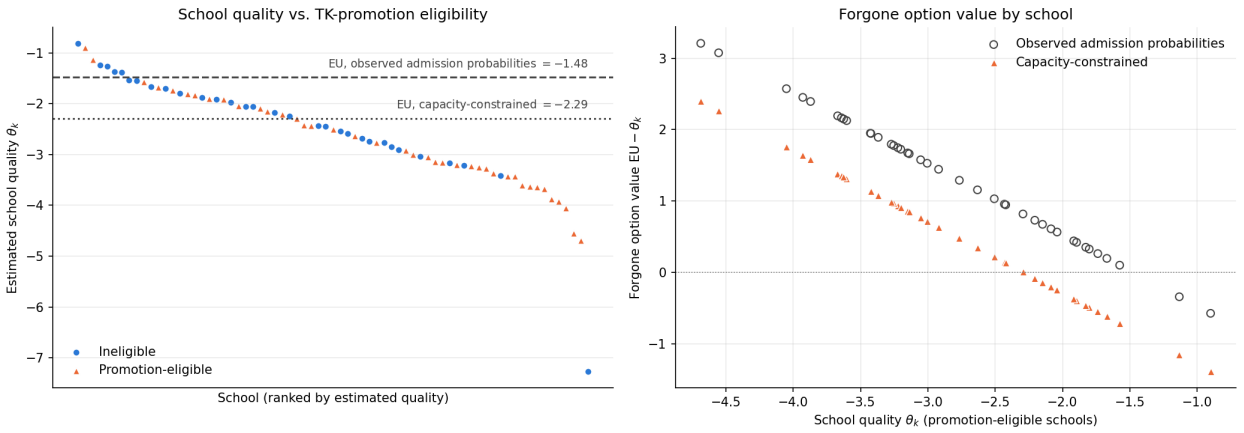


Figure 2: Left: estimated school quality θ_k (school fixed effect from the log-odds regression, 2021–22 through 2024–25), all 71 K programs ranked by quality, by TK-promotion eligibility (orange triangles: eligible; blue circles: ineligible). Horizontal lines mark the expected utility of lottery participation: dashed, the baseline formula with observed admission probabilities (EU = −1.48); dotted, the capacity-constrained simulation ($\text{EU}^{\text{cap}} = -2.29$). Schools above a line are worth more as a certain seat than the lottery is in expectation. Promotion-eligible schools are spread across the quality distribution rather than concentrated at either end, though they are somewhat overrepresented in the lowest tail, consistent with the (informal) claim that TK placements skew toward under-demanded schools. Right: forgone option value EU $-\theta_k$ (§3.6) for promotion-eligible schools, under each version of EU.

its own school with probability q and any other school k with the general probability s_{kt}. Then observed applications to k are $A_{kt} = N_t s_{kt}$, while the but-for count is

$$A_{kt}^{\text{cf}} = (N_t + P_t)\, s_{kt} + P_{kt}\,(q - s_{kt}),$$

and promotion changes applications by

$$\ln A_{kt} - \ln A_{kt}^{\text{cf}} = \underbrace{-\ln\left(1 + \frac{P_t}{N_t}\right)}_{\text{every school}} \underbrace{-\ln\left(1 + \frac{P_{kt}\,(q - s_{kt})}{(N_t + P_t)\, s_{kt}}\right)}_{\text{TK schools only}} \approx -\frac{P_t}{N_t + P_t} - q\,\frac{P_{kt}}{A_{kt}^{\text{cf}}}.$$

The first term is common to all schools: removing 601 families from a but-for cohort of 3,981 lowers applications everywhere by $\ln(3{,}981/3{,}380) = 0.16$ log points. The second term is the extra loss at a TK school: roughly the share of the school's but-for applications that its own promoted families would have supplied, scaled by q. The same algebra applies to first-choice requests, with q replaced by f, the probability a promoted family would have ranked its own school *first*. Because promoted families are a small share of any school's total requests but a large share of its first choices, the model predicts a much larger effect on first choices: under $q = f = 1$, the TK-specific effect averages −0.12 log points for all ranked requests and −0.47 for first choices across the 40 eligible schools. A difference-in-differences comparison of eligible and other schools removes the common term (absorbed by cycle effects) and estimates the TK-specific one; inverting it gives the implied q or f.

The model assumes promoted families would otherwise have behaved like other applicants apart from their attachment to their own school, and that promotion did not change how remaining families applied; the second is consistent with a strategy-proof mechanism, in which rank lists should not respond to how

	Promotion-eligible	Other schools
Schools	40	30
All ranked requests, mean per school		
2021–22 to 2024–25 average	295.4	450.1
2024–25	259.4	385.8
2025–26	215.2	335.7
2026–27	214.2	331.8
Change, 2024–25 to 2025–26	−17%	−13%
First-choice requests, mean per school		
2021–22 to 2024–25 average	45.9	70.8
2024–25	46.3	64.6
2025–26	35.1	65.8
2026–27	32.0	58.7
Change, 2024–25 to 2025–26	−24%	+2%
Promoted families per school, 2025–26	15.0	—
Promoted / 2024–25 requests (all ranks; first choice)	0.11; 0.50	—

Table 1: Kindergarten applications per school, by TK-promotion eligibility, for the 70 schools observed in every cycle from 2021–22 to 2026–27 (Mission Education Center ES, absent after 2024–25, is excluded). Counts are observed requests, without adding promoted families back. Citywide, lottery applicants fell from 3,792 in 2024–25 to 3,380 in 2025–26, while 601 families accepted promotion; ranked requests fell from 21,951 to 18,681. Source: `output/application_effects_summary.csv`.

many seats others have claimed, but need not hold in practice.

Summary statistics. Table 1 reports applications per school for the 40 promotion-eligible buildings and the 30 other kindergarten programs observed in every cycle from 2021–22 to 2026–27. Between 2024–25 and 2025–26, first-choice requests at eligible schools fell by a quarter (from 46.3 to 35.1 per school), while first choices at other schools rose slightly. All ranked requests fell at both, by 17% at eligible schools and 13% elsewhere. The promoted families were equivalent to half of an eligible school's 2024–25 first-choice requests on average, but only a tenth of its total requests. Figure 3 (left) plots the series.

Difference-in-differences. We estimate

$$\ln y_{kt} = \alpha_k + \gamma_t + \beta_{25}\, D_k \mathbb{1}[t = \text{2025–26}] + \beta_{26}\, D_k \mathbb{1}[t = \text{2026–27}] + \varepsilon_{kt},$$

where y_{kt} is all ranked or first-choice requests, D_k indicates a promotion-eligible school, and standard errors are clustered by school. The 2026–27 cycle presumably also had automatic promotion (for the 2025–26 TK class), but its counts are not in our roster, so β_{26} is reported separately and not used to calibrate the model. Table 2 reports four specifications, and Figure 3 (right) the corresponding event study over all 15 cycles.

Results. Promotion sharply reduced first-choice demand at eligible schools. The simple comparison of 2024–25 and 2025–26 gives an effect of −0.31 log points (about 27%), implying $f \approx 0.60$: most promoted families would have ranked their own TK school first. The pooled specification (2) gives a smaller effect, but the event study shows why: first choices at eligible schools were rising relative to other schools in every pre-period cycle since 2020–21 (Figure 3, left: flat at eligible schools while falling elsewhere), so averaging over the pre-period understates the break. One explanation consistent with the model is that growing TK classes were already supplying first choices to their own schools before promotion made applying unnecessary. Allowing a linear pre-trend for eligible schools (3) gives −0.38 log points and $f \approx 0.77$. PPML estimates are

	(1) 2024–25 vs. 2025–26	(2) Pooled pre-period	(3) Pooled + group trend	(4) Model intensity + trend
First-choice requests				
Eligible × 2025–26, OLS	−0.314** (0.128)	−0.176* (0.100)	−0.384*** (0.114)	—
Eligible × 2025–26, PPML	−0.294*** (0.083)	−0.195*** (0.060)	−0.346*** (0.071)	—
Model-predicted effect × 2025–26, OLS	—	—	—	0.634** (0.307)
Eligible × 2026–27, OLS	—	−0.192* (0.105)	−0.484*** (0.132)	−0.362*** (0.113)
Implied q (all ranks) or f (first choice)	0.60	0.30	0.77	—
All ranked requests				
Eligible × 2025–26, OLS	−0.067 (0.076)	−0.060 (0.065)	−0.100 (0.063)	—
Eligible × 2025–26, PPML	−0.048 (0.042)	−0.024 (0.039)	−0.073** (0.035)	—
Model-predicted effect × 2025–26, OLS	—	—	—	1.867** (0.856)
Eligible × 2026–27, OLS	—	−0.080 (0.067)	−0.135* (0.070)	−0.195*** (0.071)
Implied q (all ranks) or f (first choice)	0.52	0.47	0.80	—

Table 2: Difference-in-differences estimates of the effect of TK promotion on applications to promotion-eligible schools, relative to other schools. All columns include school and cycle fixed effects; standard errors clustered by school (70 schools) in parentheses; *, **, ***: $p < 0.10, 0.05, 0.01$. OLS rows use log counts and PPML rows use counts (coefficients in log points). Column (1) uses 2024–25 and 2025–26 only; columns (2)–(4) use 2021–22 through 2026–27, with (3) and (4) adding a linear trend for eligible schools estimated on the pre-period. Column (4) replaces the eligibility indicator with the model's predicted effect under $q = 1$ (computed from 2024–25 shares), so a coefficient of one means applications fell exactly as the model predicts if every promoted family would have ranked its own school. Implied q/f inverts the model's predicted difference-in-differences at the OLS estimate. Source: `output/application_effects_did.csv`, `output/application_effects_model.csv`.

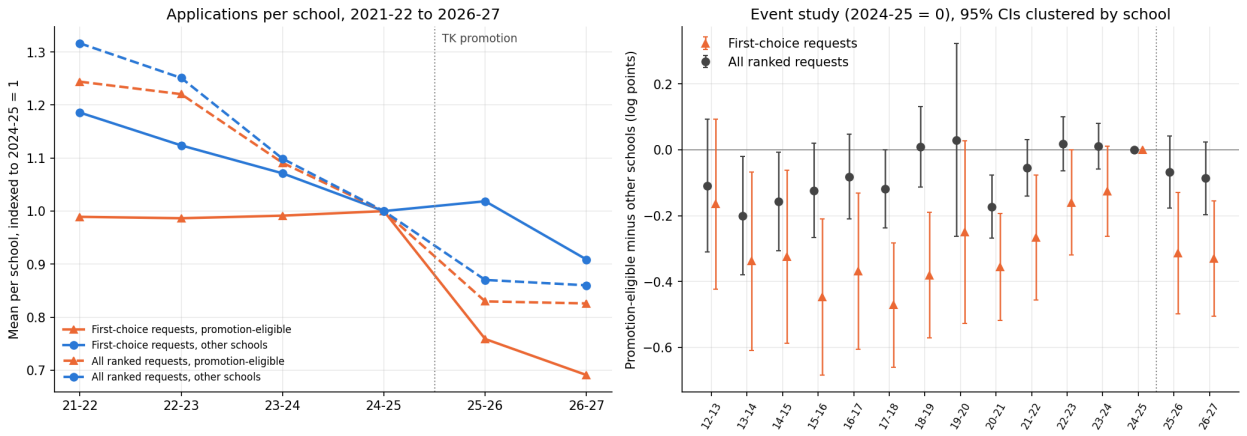


Figure 3: Left: mean requests per school, indexed to 2024–25, for promotion-eligible schools (orange triangles) and other schools (blue circles); solid lines are first-choice requests and dashed lines all ranked requests. Right: event-study coefficients on eligible × cycle (log points, 2024–25 = 0) for first-choice and all ranked requests, with 95% confidence intervals clustered by school. Source: `output/application_effects_event_study.csv`.

similar. We maintain $f = 1$ throughout: a family that accepts a guaranteed seat at its own school would have ranked that school first had it applied. Figure 1 is the direct evidence—adding the promoted families back more than accounts for the fall in first-choice requests at eligible schools. The difference-in-differences estimates above imply somewhat less, and we read that gap as reflecting the pre-trend the event study shows rather than as evidence that promoted families would have ranked elsewhere. We distinguish two objects. Whether a family would have *ranked its own school first* is not the same as whether it *prefers that school to the lottery*: accepting promotion reveals the second directly, and implies the first only because a family holding top priority that ranks its school first is assured of a seat there, which makes ranking first and accepting equivalent choices in this setting. That equivalence would fail if submitted lists were length-constrained or if priority were uncertain. The distinction has teeth here, because the sign of the first-choice comparison depends on it: adding promoted families back raises eligible schools' first-choice count relative to other schools only if f exceeds 0.80, and every f the difference-in-differences implies is below that. The reversal reported above therefore rests on the maintained $f = 1$, not on the estimated f. The effect persists into 2026–27, consistent with promotion continuing.

Effects on all ranked requests are smaller and imprecise (-0.06 to -0.10 log points, none significant at 5% by OLS), as the model predicts: promoted families were a tenth of an eligible school's total requests, so even $q = 1$ implies only a 0.12 log-point TK-specific drop. The implied q ranges from 0.47 to 0.80 across specifications, with confidence intervals spanning both zero and one. Column (4) offers a sharper test of the model's cross-school prediction, regressing on each school's model-predicted effect: schools where promotion should have mattered more did lose more. For first choices the coefficient is 0.63 (OLS; 0.74 PPML), of the same order as the implied f from columns (1)–(3); its confidence interval includes the 1 that $f = 1$ predicts, as does that of the all-ranks coefficient below. For all ranked requests it is 1.87 (s.e. 0.86), above one: high-intensity schools lost more total requests than their promoted families alone account for. That would be consistent with remaining families ranking schools less often when promotion had visibly claimed many of their seats, but the estimate is imprecise and the confidence interval includes one.

We now turn to a structural model of why families decline the lottery, built on an expected-value calculation over the estimated θ_k.

3.6 The cost of application

Our quality estimates θ_k measure the value of *applying* to a school, not of attending it. A family that accepts automatic promotion attends its TK school with certainty; a family that enters the lottery attends its k-th choice only if every higher-ranked school rejects it. Valuing lottery participation therefore requires admission probabilities.

Expected value of the lottery. A family that ranks schools by preference and is assigned the highest-ranked school that admits it has expected payoff

$$\mathrm{EU} = \sum_{k=1}^{K} \theta_k \, P_k \prod_{j \prec k} (1 - P_j) \,,$$

where schools are indexed $1, \ldots, K$ in decreasing order of θ (truthful preference order, given SFUSD's mechanism is strategy-proof), $j \prec k$ denotes every school ranked above k, and P_k is school k's admission probability. This is exactly $P(\text{admit}_1)\theta_1 + P(\text{admit}_2)P(\text{reject}_1)\theta_2 + \cdots$: a family only reaches its k-th choice if it was

rejected everywhere it ranked higher. We treat rejection as independent across schools—a simplification forced by not observing individual-level match outcomes, which would let rejections correlate (e.g. through a common lottery number) in ways we cannot identify here.

P_k itself we proxy with $\min(1, \text{capacity}_k/\text{demand}_k)$: demand_k is school k's average total (all-rank) lottery request volume over the estimation window, and capacity_k is its actual kindergarten enrollment, averaged over the two most recent CDE school-level enrollment files (2023–24, 2024–25)—occupied seats being the closest available proxy for how many seats exist. Matching lottery school names to CDE school names (by hand where needed) gives a capacity for all 71 schools in the window, and admission probabilities from 5.5% (SF Public Montessori) and 7.5% (Clarendon ES) to a median of 17%, with only Mission Education Center ES undersubscribed. (The full panel adds 15 standalone TK program sites that appear in early cycles' kindergarten listings, with no separately reported kindergarten enrollment; they are not in the main window, and in the full-panel robustness check they fall back to $P_k = 1$.) Code and the full crosswalk are in `src/capacity.py`.

Plugging in gives EU $= -1.48$, below Clarendon ES's θ of -0.81 (the best K school in the district by this measure), as it must be once admission is uncertain.

Forgone option value. A family offered automatic promotion into school k gives up $\text{EU} - \theta_k$ by accepting. Averaged over the 601 families who accepted (weighting each school by its acceptances), this is 1.24 utility units, comparable to the θ-gap between Clarendon ES and Sunnyside ES (#24 of 71). Under the model of §4 it is the lottery distaste of a family exactly indifferent between promotion and lottery; that section uses observed take-up to move from this threshold to an average.

Two schools—West Portal ES and Alvarado ES, the highest-θ promotion-eligible schools (#2 and #3 of 71)—have *negative* forgone option value: a family zoned there is better off taking the certain seat than gambling on the lottery, admission odds included, even with no distaste for the lottery at all. Families at these two schools who nonetheless entered the lottery were, by this model, taking a negative-expected-value bet (heterogeneous beliefs, an idiosyncratic preference draw we do not observe, or model misspecification are all possible). Figure 2 marks EU against the distribution of θ_k (left panel) and plots the measure by school (right panel).

This is still not the dollar-denominated answer the introduction wants, and the independence assumption above is a real one (true deferred-acceptance rejections likely correlate through priorities and lottery numbers in ways that would push EU further down, since correlated rejection reduces the value of a long list relative to independent draws). The next paragraph addresses a larger problem with the same formula.

A capacity-constrained simulation. The formula above gives a single representative family an independent admission draw at every school, so that family collects a ticket at each of the most popular schools. Under it, a family lands in one of the top 8 schools with probability 0.52 and one of the top 15 with probability 0.78; but the top 8 schools have roughly 630 kindergarten seats for roughly 4,000 applicants, so at most about 16% of families can. Summed over families, the formula fills the best schools many times over, which is why only 7 of 71 schools have θ_k above the baseline EU.

We therefore also simulate the 2025–26 lottery for the full cohort under the model of §3, with seat limits. The cohort is the 3,981 families who would have applied absent the opt-out (lottery applicants plus the 601 who accepted promotion). Each family draws $u_{ik} = \theta_k + \varepsilon_{ik}$ and an outside option $\xi + \varepsilon_{i0}$, and ranks every school it prefers to the outside option; ξ is calibrated so the mean list length matches the 4.84

requests per applicant the panel implies for the but-for cohort. Seats are each school's CDE kindergarten enrollment, 3,642 in total. Seats are assigned by random serial dictatorship—one lottery number per family, no priorities, since SFUSD's sibling, attendance-area, and CTIP1 priorities are unobserved without choice data—and families placed nowhere on their list are assigned administratively to a random remaining seat, as SFUSD does. In the simulation, 77% of families are placed from their list, 14% administratively, and 9% find no seat left (CDE enrollment counts occupied seats, so it understates true capacity).

We value the outcome by the θ of the assigned school, the common-value component, so the result is directly comparable to θ_k of the certain TK seat. This gives $\text{EU}^{\text{cap}} = -2.29$, against -1.48 for the baseline formula. Because seats are scarce in aggregate, every seat is filled, and EU^{cap} is exactly the seat-weighted average of θ_k (it does not vary across simulation draws): in expectation, a family entering an oversubscribed lottery receives the average seat. Heterogeneous preferences determine *who* gets which seat, but not the average θ of the seats handed out. They do matter for the full utility $\theta_k + \varepsilon_{ik}$ of the assignment, which we set aside here because the same idiosyncratic term attaches to the certain TK seat (see §4).

Against EU^{cap}, 30 of 71 schools have θ_k above the lottery's expected value, and the family-weighted forgone option value falls to 0.43 (about the θ-gap between Clarendon ES and Lawton K-8, #4 of 71). Under the simulation, 13 promotion-eligible schools, accounting for 227 of the 601 acceptances, now have negative forgone option value: for those families, accepting promotion is the better bet even with no distaste for the lottery. Figure 2 shows both versions of EU and of the forgone option value. We take the capacity-constrained simulation as our preferred measure of the lottery's value, and report the baseline formula alongside it as an upper bound on that value. The baseline places a family in one of the top 8 schools with probability 0.52 when at most about 16% of families can be, so it overstates what the lottery delivers and therefore overstates distaste. The simulation errs in the other direction: it ignores priorities, which concentrate popular seats among families with sibling or attendance-area claims, so a family without them does somewhat worse than the average seat. Both value the lottery over the full set of schools rather than over the list a family actually submits; because observed list lengths are variable and only coarsely reported beyond the ninth rank (§2), recomputing EU over the top n ranks for a range of n is a natural robustness check we leave for the individual data.

Per-school results, admission probabilities, and the CDE-matching crosswalk are in `output/lottery_distaste_estimates` and `output/school_admission_probabilities.csv`; `output/lottery_distaste_summary.csv` carries the summary statistics, and the estimation code lives in `run_distaste_analysis.py` and `src/`.

4 Promotion rates

Not every eligible family accepted promotion: 601 of 820 (73%) did. As described in §3, the roster's two-program buildings (DeAvila, Huerta) are combined, giving 40 buildings. The roster's eligibility count matches each school's 2024–25 TK enrollment in CDE's census-day file to within two students at 29 of the 40 schools, with the largest discrepancy 6 students, so the school-level ratio $r_k = \text{accepted}_k/\text{eligible}_k$ is exactly the share of last year's TK class that took the guaranteed seat. Rates range from 41% (Longfellow, Flynn, Chavez) to 100% (Key); per-school detail is in `output/promotion_rates.csv`.

Promotion rates and school quality. If a guaranteed seat is worth more at a better school, promotion rates should rise with school quality. They do not (Figure 4). Across the 40 promotion-eligible buildings, the correlation between r_k and our demand-based θ_k is -0.05; with SBAC 2023–24 percent met-or-exceeded it is

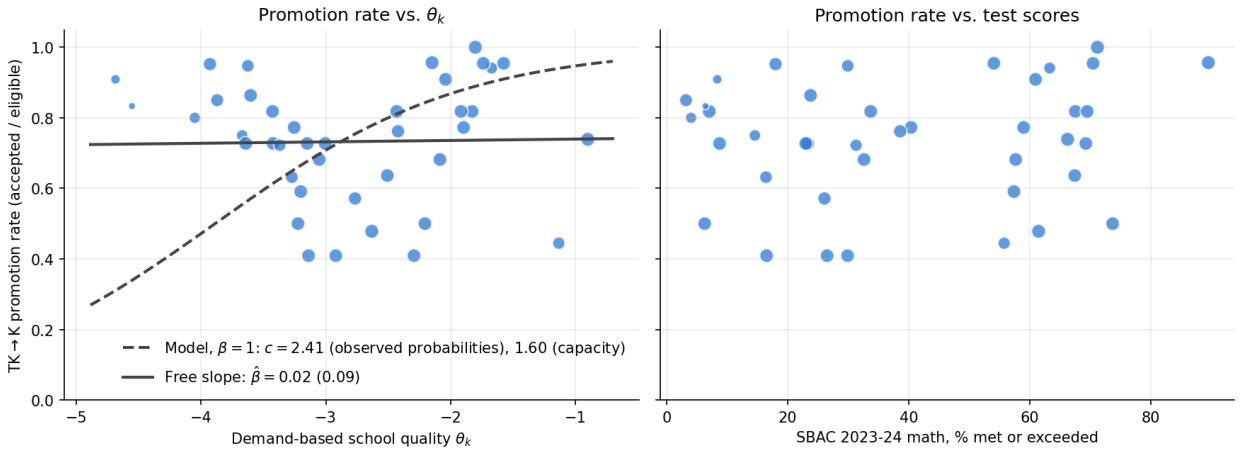


Figure 4: TK→K promotion rate by school against demand-based quality θ_k (left) and SBAC 2023–24 math proficiency (right); marker area is proportional to the number of eligible TK students. Left: the dashed curve is the model-implied acceptance probability $\Lambda(\theta_k - \mathrm{EU} + c)$ at the pooled estimate ($c = 2.41$ with the baseline EU, or equivalently $c = 1.60$ with $\mathrm{EU}^{\mathrm{cap}}$); the solid curve frees the coefficient on $\theta_k - \mathrm{EU}$.

0.16 (ELA) and 0.12 (math), none distinguishable from zero. The only covariate with a hint of a relationship is the school's unduplicated-pupil (low-income/EL/foster) share, at -0.23 ($p \approx 0.15$). Since θ_k, test scores, and demographics are themselves highly collinear across these schools (pairwise $|\rho|$ from 0.68 to 0.97), there is no room to say more with 40 observations.

Using the rate to estimate distaste. The probability-weighted measure of §3.6, $\mathrm{EU} - \theta_k$, is the option value a family forgoes by accepting promotion. It does not use the promotion rate at all: it is computed identically whether 10% or 100% of a school's TK families accepted. Under the model, it is the lottery distaste of the *marginal* family, the one indifferent between promotion and lottery. Inverting the acceptance probability of the model section gives, school by school,

$$c_k = \underbrace{\ln \frac{r_k}{1 - r_k}}_{\text{logit of promotion rate}} + \underbrace{(\mathrm{EU} - \theta_k)}_{\text{forgone option value}} .$$

A promotion rate below one does mean families find the lottery less distasteful than they would if everyone had accepted (as $r_k \to 1$, $c_k \to \infty$), but the relevant comparison is with the option-value measure, which implicitly sets $r_k = 1/2$. Since 73% of families accepted, the typical family's distaste is *above* the marginal family's, and incorporating the rate raises the estimate. Fitting a single c by grouped-logit maximum likelihood across all 820 eligible families, against the preferred capacity-constrained $\mathrm{EU}^{\mathrm{cap}}$, gives $\hat{c} = 1.60$ (s.e. 0.08, which ignores sampling error in θ_k and EU and is therefore too small), against 0.43 for the forgone option value. On the scale of §3.6, 1.60 is roughly the θ-gap between Clarendon ES and Lakeshore ES (#32 of 71), close to the gap between the best and the median school in the district. Using the baseline EU instead gives the larger $\hat{c} = 2.41$ (s.e. 0.08; against a forgone option value of 1.24), roughly the gap between Clarendon ES and Bryant ES (#55). With the slope restricted to one, the data identify only $c - \mathrm{EU}$, so the two estimates differ by exactly the gap between the two expected values, 0.81, and trace the same acceptance curve; the choice between them is a choice of EU, not something the promotion data can settle.

The slope restriction. The model ties the slope of acceptance in θ_k to one: a family at a school one utility unit better should be Λ' more likely to accept. Freeing that coefficient gives $\hat{\beta} = 0.02$ (s.e. 0.09); the restriction $\beta = 1$ is firmly rejected. The data indicate that promotion decisions are close to unrelated to how much the city wants to attend the school in question. Three readings, which are not mutually exclusive:

- *Selection on match value.* Families at a low-θ school who chose it for TK value it more than the city does (z_{ik} and $x_i'\beta_k$ are doing the work); this is exactly what individual data would let us test.
- *Learning.* A year at the school replaces the citywide prior θ_k with first-hand information, so θ_k is the wrong index for TK families.
- *Non-acceptance is not lottery entry.* Some non-accepting families leave SFUSD (private, charter, out of city) rather than entering the lottery. Those are rejections of the school, not a taste for the lottery, and they bias $\hat{c}$ *down*.

Under any of these, $\hat{c}$ (1.60, or 2.41 under the baseline EU) is best read as the average certainty premium implied by a 73% take-up rate, rather than a structural estimate validated by cross-school variation. The directions matter as much as the readings. Every channel above except the last pushes $\hat{c}$ *up*: both selection on match value and learning make a family's own school worth more to that family than θ_k says, so part of what we attribute to distaste for the lottery is in fact taste for the school. Only families who leave SFUSD rather than enter the lottery push the other way, and that channel is the one of the three that could be bounded with data we have asked for: October enrollment would show how many of the 219 non-accepting families never enrolled in an SFUSD kindergarten. On balance we therefore read $\hat{c}$ as an upper bound on distaste for lottery participation, subject to that one offset. The individual data are what would separate these channels.

5 Measuring school quality over time

The model treats school quality θ_k as fixed, estimated by pooling request data over the main window. Two checks address whether that matters.

Re-estimating with other windows, and year by year. Table 3 re-runs the full chain—admission probabilities, both expected values, forgone option value, certainty premium, and the free slope—with θ_k pooled over the full 15-year panel, pooled over the main window extended through the 2025–26 treatment year, and measured separately in each year of the main window. For a single year t, θ_{kt} is school k's log-odds of being ranked that year alone, and admission probabilities use that year's demand alone. The expected values themselves shift with each specification's normalization, so the table reports only the quantities that depend on gaps between schools. Across all seven rows, the forgone option value stays between 1.17 and 1.30 (baseline) and 0.42 and 0.47 (capacity-constrained), the certainty premium between 2.32 and 2.47 (baseline) and 1.56 and 1.66 (capacity-constrained), and the free slope is never distinguishable from zero.

Stability of school rankings. Figure 5 reports the Spearman rank correlation of single-year quality θ_{kt} between every pair of the 15 application cycles (left), and between each cycle and the main specification's pooled θ_k (right). Rankings are very stable from year to year: adjacent cycles correlate at 0.95 to 0.99. They drift slowly over longer spans, falling to 0.86 between 2012–13 and 2026–27. 2019–20 stands out as the

Quality measure	Forgone option value baseline	Forgone option value capacity	Certainty premium $\hat{c}$ baseline	Certainty premium $\hat{c}$ capacity	Free slope $\hat{\beta}$	Corr. (r_k, θ_k)
Main: pooled, 2021–22 to 2024–25	1.24	0.43	2.41	1.60	0.02	−0.05
Pooled, full panel (2012–13 to 2026–27)	1.22	0.47	2.41	1.66	0.04	−0.03
Pooled, 2021–22 to 2025–26	1.26	0.43	2.43	1.59	0.02	−0.06
Single year, 2021–22	1.25	0.44	2.44	1.63	0.07	−0.01
Single year, 2022–23	1.17	0.42	2.32	1.56	0.01	−0.06
Single year, 2023–24	1.27	0.43	2.43	1.60	0.02	−0.05
Single year, 2024–25	1.30	0.43	2.47	1.61	−0.01	−0.08

Table 3: Distaste estimates under alternative measures of school quality. "Pooled" rows estimate one θ_k per school from school and year fixed effects over the stated cycles; "single year" rows use that cycle's log-odds alone. Forgone option values are family-weighted over the 601 acceptances. The free slope's standard error is 0.09–0.10 in every row. Source: `output/robustness_quality_measurement.csv`.

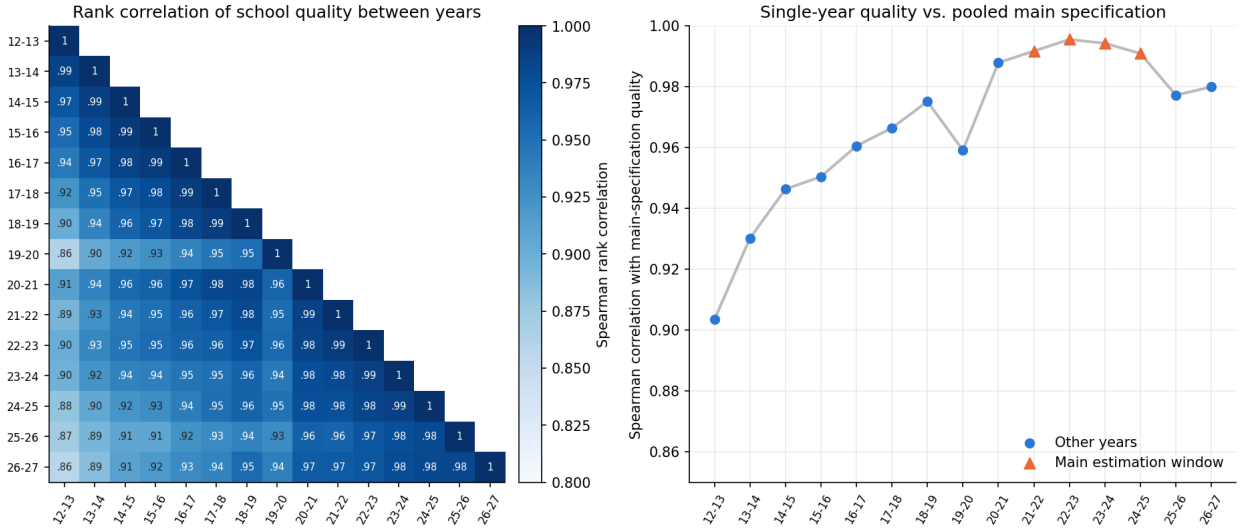


Figure 5: Left: Spearman rank correlation of single-year school quality θ_{kt} between each pair of application cycles, over schools present in both (65–86 schools per cycle). Right: each cycle's rank correlation with the main specification's pooled θ_k (2021–22 through 2024–25, orange triangles). Source: `output/quality_rank_correlation_by_year.csv`, `output/quality_rank_correlation_with_main.csv`.

noisiest cycle (correlation 0.96 with the main specification, against 0.98–0.99 for its neighbors). Within the main window every cycle correlates at 0.99 or above with the pooled estimate, and the post-decision cycles (2025–26, 2026–27) at 0.98. A handful of schools did move: relative to the full panel, 10 of the 71 schools shift by more than 0.2 utility units in the recent window, the largest rise being Muir (John) ES at 0.38 and the largest fall Grattan ES at 0.28; these shifts are the reason we prefer the recent window. School-specific trends will be estimable directly once we have individual choice data.

6 Other concerns

- Learning (by being at the school)
- Endowment effect
- Selection (people who know they want to go to the school)

- The fact that, implicitly, measured preferences based on rank applications are net of acceptance probability

A Estimating risk aversion directly

The certainty premium c of §4 is a reduced-form "distaste" added to a risk-neutral comparison. An alternative reading of the same choices is that families are risk-averse over school quality: they prefer the certain seat because the lottery is risky, not because lottery participation is costly in itself. This appendix estimates that risk aversion directly and compares it to canonical values.

Setup. Each of the 820 eligible families makes one binary choice between a sure outcome, kindergarten at its own TK school ($s_i = \theta_{k(i)}$), and a lottery L over the θ of the school the kindergarten lottery would assign it. Without individual data we treat these as 820 choices by a single representative agent who balances the guaranteed payoff of staying against the uncertain lottery. All families face the same lottery, and the sure outcome varies across the 40 buildings, giving 40 choice "designs." We use both lottery definitions from §3.6: the baseline, in which school k is reached with probability $P_k \prod_{j \prec k}(1 - P_j)$, and the capacity-constrained simulation, in which L is the distribution of assigned schools among families who receive a seat (the 9% with no seat are excluded, which if anything understates the lottery's downside).

We use constant absolute risk aversion (CARA), $u(x; r) = (1 - e^{-rx})/r$, with r measured per unit of θ. CRRA is unsuitable here: θ takes negative values and has no natural zero (each specification normalizes it differently), whereas CARA is invariant to adding a constant to every outcome, so r does not depend on the normalization. θ's scale is pinned by the unit-scale logit shocks, so r is well defined. Code is in `src/risk_aversion.py`.

Representative agent. Following the noise-in-certainty-equivalent specification (which, unlike utility-space noise, keeps choice probabilities monotone in r; Apesteguia and Ballester 2018), the family chooses the lottery with probability

$$\Pr(\text{lottery} \mid k) = \Lambda\left(\frac{\mathrm{CE}_L(r) - \theta_k}{\mu}\right), \qquad \mathrm{CE}_L(r) = -\frac{1}{r}\ln \mathbb{E}_L\left[e^{-r\theta}\right].$$

Setting $\mu = 1$, the scale of the logit shocks that already normalizes θ in the promotion model, this is exactly the acceptance model of §4 with the certainty premium replaced by curvature: $\mathrm{CE}_L(\hat{r}) = \mathrm{EU} - \hat{c} = -3.89$ under either lottery definition. The estimates are $\hat{r} = 5.0$ (robust s.e. 0.14) for the baseline lottery and $\hat{r} = 6.3$ (robust s.e. 0.82) for the capacity-constrained lottery (Figure 6, left); the implied risk premia, $\mathbb{E}[L] - \mathrm{CE}_L$, equal $\hat{c}$ (2.41 and 1.60). Like $\hat{c}$, these standard errors ignore estimation error in θ and the admission probabilities. With these data, a risk-neutral family with distaste c and a CARA family with $r = \hat{r}$ make identical choices: both explain the same single moment, the 73% take-up rate. Individual data would help separate them, because families with different priorities face lotteries with different spreads, and risk aversion responds to the spread while a fixed distaste does not. The separation is not automatic, however: priority is a function of where a family lives and whether it has a sibling enrolled, and those same variables plausibly shift the utility of the school itself, so identification would rest on an exclusion restriction—some determinant of the spread a family faces that does not enter its preferences directly—which we do not have and would need to argue for.

What the data cannot identify. Freeing μ fails: the likelihood improves without limit toward $r \to \infty$, and the guard in the code reports non-identification. This is the flat promotion-rate gradient of §4 again: the slope of acceptance in θ_k is $1/\mu$, and it is indistinguishable from zero. The utility-space (Fechner) and range-normalized (contextual; Wilcox 2011) noise specifications likewise run to a parameter bound. A heterogeneous-agent model, in which each family has its own $r_i \sim N(m, \sigma^2)$ and chooses the lottery exactly when r_i is below its school's indifference point r_k^* (where $\mathrm{CE}_L(r_k^*) = \theta_k$), implies $\Pr(\text{lottery} \mid k) = \Phi((r_k^* - m)/\sigma)$, a probit of choice on r_k^*. The probit is estimated over school-level designs rather than the 820 families directly: 40 designs for the baseline lottery, and 39 for the capacity-constrained one, where 11 families at one school are dropped because their design is dominated: that school's own θ_k lies at or below the worst outcome the capacity-constrained lottery can deliver, so the lottery weakly dominates the certain seat for *every* r_i and the choice carries no information about m or σ. The share choosing the lottery should therefore rise with r_k^*; it is flat (Figure 6, right; probit slope -0.004, s.e. 0.008, for the baseline lottery and -0.002, s.e. 0.005, for the capacity-constrained one). Taken literally, the 95% confidence bound on the slope requires a dispersion of risk aversion σ of at least 79 (baseline) or 115 (capacity-constrained) per θ unit, an order of magnitude beyond the representative estimate; we read this as a rejection of the pure-heterogeneity model rather than as an estimate. With one choice per family, within-family noise and between-family heterogeneity cannot be separated: the representative-agent model attributes all variation to noise, and the heterogeneous model attributes all of it to heterogeneity.

Fragility in the tail. Matching a certainty equivalent of -3.89 requires the lottery's worst outcomes to carry nearly all the weight. Under the baseline lottery, the probability of being assigned a school with θ below -3.89 is about 5×10^{-7}; under the capacity-constrained lottery it is 2.8%. Accordingly, $\hat{r}$ is driven by the lowest few schools: dropping the single lowest outcome and renormalizing moves $\hat{r}$ from 5.0 to 21 (baseline) and from 6.3 to 8.0 (capacity-constrained); dropping the lowest three moves both above 100. Expected-utility curvature can rationalize 73% take-up only by letting a few-percent (or vanishing) chance of the district's least-demanded schools dominate the decision. That pattern is better described by overweighting of small probabilities or worst-case reasoning (ambiguity aversion) than by smooth risk aversion, and is consistent with the probability distortions found in field insurance choices (Barseghyan et al. 2013).

Comparison to canonical risk aversion. Canonical estimates are of relative risk aversion over money, so a comparison needs either a unit-free summary or a conversion from θ to dollars. We report both (Table 4).

- *Unit-free.* The risk premium as a share of its maximum, $(\mathbb{E}[L] - \mathrm{CE}_L)/(\mathbb{E}[L] - \min L)$, is 0.42 (baseline) and 0.67 (capacity-constrained). On Holt and Laury's (2002) high-risk lottery at the 50/50 row (\$3.85 or \$0.10), the CRRA coefficients producing the same shares are 0.57 (baseline) and 0.91 (capacity-constrained). The baseline figure sits inside the 0.3–0.7 range typical of small-stakes laboratory choices, while the preferred capacity-constrained figure sits above it (Holt and Laury 2002; Harrison and Rutström 2008). Measured against the lottery's own range, then, families are about as risk-averse as lab subjects—but the share depends on $\min L$, which is precisely the fragile tail above, so we do not read this agreement as support for the CARA reading.
- *In dollars.* If one unit of θ is worth D dollars and the relevant wealth is W, relative risk aversion is $\rho = (\hat{r}/D)W$. Taking $W =$ \$1 million as an illustrative San Francisco family's wealth, and $D =$ \$50,000 (one θ unit is roughly the gap between the district's most-demanded school and its 16th), gives $\rho \approx 100$ (baseline) to 125 (capacity-constrained). For ρ to fall to 2, the upper end of the macro and labor-supply

	Baseline lottery	Capacity-constrained
CARA $\hat{r}$ (per θ unit), $\mu = 1$	5.01 (0.14)	6.27 (0.82)
Lottery mean / s.d.	−1.48 / 0.40	−2.29 / 0.82
Certainty equivalent	−3.89	−3.89
Risk premium ($= \hat{c}$)	2.41	1.60
Risk premium / lottery s.d.	6.04	1.96
Probability of an outcome below the CE	5×10^{-7}	0.028
$\hat{r}$ after dropping lowest 1 / 3 outcomes	21 / 100	8.0 / 107
Premium as share of maximum	0.42	0.67
Holt–Laury-equivalent CRRA ρ	0.57	0.91
Implied CRRA if θ unit = \$50k, W = \$1M	100	125
\$ per θ unit needed for $\rho = 2$, W = \$1M	$\$2.5M$	$\$3.1M$
Model A (heterogeneous r_i): probit slope on r_k^*	−0.004 (0.008)	−0.002 (0.005)

Table 4: Risk aversion implied by the TK promotion decision, and comparisons to canonical values. Robust standard errors in parentheses. Source: `output/risk_aversion_estimates.csv`, `output/risk_aversion_canonical_comparison.csv`.

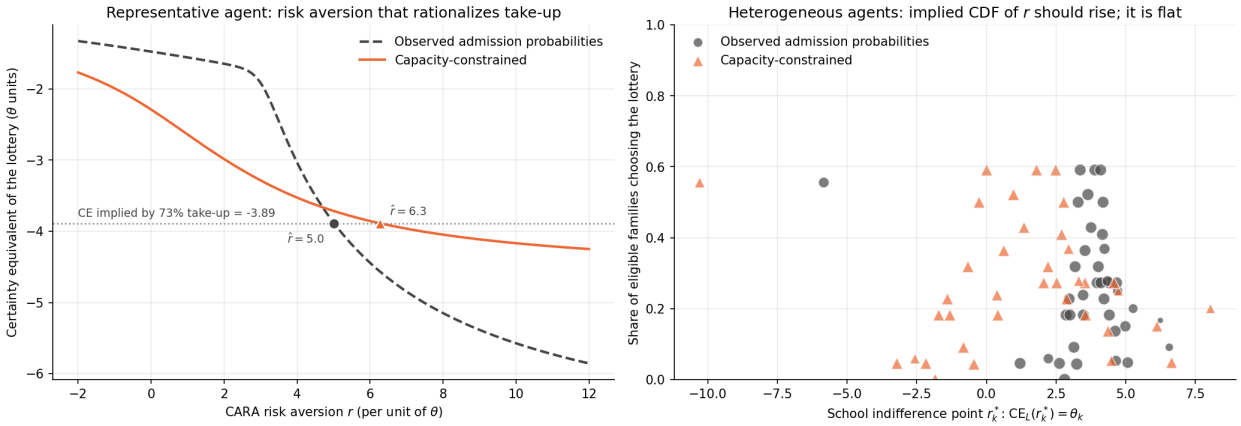


Figure 6: Left: certainty equivalent of the lottery as a function of CARA risk aversion r, for each lottery definition; the dotted line is the certainty equivalent implied by the 73% take-up rate, and markers are the estimates. Right: for each school, the share of eligible families choosing the lottery against the school's indifference point r_k^* (marker area proportional to eligible families). Under heterogeneous risk aversion these points trace the CDF of r and should rise from left to right.

range (Chetty 2006), a single θ unit would have to be worth \$2.5M–\$3.1M. These magnitudes resemble the implausibly high coefficients implied by insurance deductible choices (Sydnor 2010), and Rabin's (2000) calibration argument applies: no reasonable curvature over lifetime wealth produces premia this large on stakes this size. We do not yet have a θ-to-dollar conversion of our own; the distance coefficient from the individual data will supply one.

Validation. Before running on the data, `validate_risk_aversion.py` checks the estimators on simulated data (30 checks, all passing; `output/risk_aversion_validation.csv`). The heterogeneous-agent probit recovers a known CRRA distribution $N(0.5, 0.3^2)$ over 200 simulated samples of 2,000 (mean estimates 0.501 and 0.300, 95% interval coverage 0.94 and 0.94); the representative-agent estimator recovers known (r, μ) under all three noise specifications; and the guards correctly refuse to report m, σ, or a free-μ estimate when only one design is observed, and drop dominated designs.

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